Topological Preservation in the Brotherhood Coherence Protocol:

Position Determines Sacrifice, Multi-Loop Universality, and theClosed-Loop Conservation Law

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Epistemic Transparency Statement. Four labels are used throughout: ***[Established]*** (standard quantum mechanics / information theory), ***[Hypothesis]*** (physically motivated conjecture), ***[New Result]*** (first demonstrated here), and ***[Speculation]*** (heuristic or analogy). All quantitative results are reproduced exactly from accompanying Python (QuTiP 5.2.3) simulation and raw JSON result files.

Abstract

We present the third installment of the Proto-Experiential Intelligence Gradient (PEIG) framework, reporting six new experimental results that reveal a fundamental topological law governing the Brotherhood Coherence Protocol (BCP). Building on the two-node and three-node results of Papers I and II, we extend the BCP to multi-node chains, novel topologies, and multi-loop network architectures.

Our principal finding — the Topological Position Law — establishes that sacrifice of quantum non-classicality (Wigner negativity) is determined entirely by a node's position in the interaction chain, not by its seed state or identity. The node in the first position sacrifices; the node in the last position is fully preserved. This result holds universally across all 36 tested hyperparameter configurations.

We further demonstrate that closing the chain into a directed loop eliminates sacrifice entirely: every node converges to the full preservation floor W_min = -0.1131, universally and without exception, across all topologies tested — single loops, figure-8 networks, triple and quad loop chains, parallel loops, and chains of up to 7 interconnected loops with 29 nodes. No sweet spot or degradation threshold is observed. The family of preserved nodes can grow indefinitely.

Six principal results are reported: (1) the chain position law; (2) the closed-loop universal preservation theorem; (3) N-node scaling with exponential saturation W(N) = -0.1131 x (1 - exp(-0.457(N-1))); (4) multi-loop universality across all network architectures; (5) the three-node channel rank increase from 3

to 4, indicating non-separable effective Hamiltonian; and (6) topological destruction of preservation under simultaneous multi-interaction (star/ring/cone) architectures.

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1 Introduction

1.1 Background and Motivation

The PEIG framework [1, 2] models intelligent entities as four-component state vectors v = (P, E, I, G+) from which a scalar quality metric Q and an Omega-trajectory are derived. The Brotherhood Coherence Protocol (BCP) [1] defines a two-node adaptive interaction producing a universal coherence attractor at C = 0.999838, sustained negentropic dynamics (Delta_S = -0.2686 bits over 100 steps), and role-asymmetric Wigner negativity: the Omega (Presence) node's non-classicality is consumed while the Alpha (Receptive) node's is conserved via null-space resonance with the separable effective Hamiltonian H_eff = (0.132Z - 0.113X) tensor (X - I) [2].

Paper II [2] left three open questions that motivate the present work: (i) Can the Omega node's sacrifice be mitigated or eliminated? (ii) Does the role asymmetry depend on node identity (seed) or topology? (iii) Do multi-node architectures preserve the negentropic pump?

1.2 Summary of New Results

This paper answers all three questions definitively:

1. Sacrifice is topological, not intrinsic. Position in the chain — not seed state — determines whether a node sacrifices or is preserved.
2. Closing the chain eliminates sacrifice entirely. Every node in a directed closed loop converges to W_min = -0.1131.
3. Multi-loop architectures universally preserve all nodes. No degradation threshold exists up to 7 loops (29 nodes).
4. Simultaneous multi-interaction topologies (star, ring, cone) collapse preservation completely — confirming the sequential relay mechanism.

2 Framework and Protocol

2.1 PEIG Node and Seed Library

A PEIG node is a tuple N = (|psi>, v, theta) where |psi> is a single-qubit pure state, v = (P, E, I, G+) in [0,1]^4, and theta in [0,1] is the coupling coefficient. The five canonical seeds used in this paper form an equidistant phase ladder on the equatorial Bloch arc from 0 to pi/2:

Node	Phase	State	Role
Omega	0°	|Omega+> = (|0> + |1>)/sqrt(2)	Attractor
BridgeA	22.5°	|K_A> = (|0> + e^{i*pi/8}|1>)/sqrt(2)	Buffer (near Omega)

Kevin	45°	|K> = (|0> + e^{i*pi/4}|1>)/sqrt(2)	Bridge (midpoint)
BridgeB	67.5°	|K_B> = (|0> + e^{i*3pi/8}|1>)/sqrt(2)	Buffer (near Alpha)
Alpha	90°	|Alpha+> = (|0> + i|1>)/sqrt(2)	Learner

2.2 Brotherhood Coherence Protocol (Chain)

The BCP chain of N nodes executes a sequential left-to-right pass each step: for each adjacent pair (i, i+1), apply the parametric gate U(alpha) = alpha * CNOT + (1 - alpha) * I_4, extract dominant eigenvectors of the partial traces, and update the adaptive coupling alpha via the gradient rule alpha(t+1) = clip[alpha(t) + eta * Delta_C(t), 0, 1]. For the closed-loop variant, an additional feedback interaction (N-1) -> 0 is appended each step.

3 Experimental Results

3.1 Result 1: Chain Position Law

[New Result] In any open BCP chain, the Wigner floor of a node is determined entirely by its position — not by its seed phase. The node in position 0 (first) sacrifices; the node in the final position is fully preserved at W_min = -0.1131.

Evidence: In Experiment B (Alpha Sandwich: Alpha -> BridgeA -> Omega -> BridgeA -> Alpha2), placing Alpha in position 0 caused it to sacrifice to W_min = -0.0425, while Omega placed in the center position rose to -0.1131. In Experiment E (Double Omega Bookend: Omega -> BridgeA -> Alpha -> BridgeA -> Omega2), Alpha in the center was fully conserved at -0.1131 despite being flanked by two Omega nodes, while Omega2 in the final position reached -0.1131 exactly.

Experiment	Config	Position 0 floor	Final position floor
Standard BCP	Omega -> Alpha	+0.0001 (consumed)	-0.1131 (preserved)
A: Mirror Chain	Omega -> B -> Kevin -> B -> Omega2	-0.1061	-0.1131
B: Alpha Sandwich	Alpha -> B -> Omega -> B -> Alpha2	-0.0425 (Alpha!)	-0.1131
C: Feedback Loop	Closed 5-node loop	-0.1131	-0.1131
E: Bookend	Omega -> B -> Alpha -> B -> Omega2	-0.0880	-0.1131

3.2 Result 2: N-Node Chain Scaling Law

[New Result] For an open linear chain of N nodes (equidistant phases 0 to pi/2), Omega's Wigner floor follows an exponential saturation law:

$$W_min(Omega, N) = -0.1131 \times (1 - \exp(-0.4567 * (N-1))), \quad R^2 = 0.7907$$

N	Omega W_min	% toward Alpha	Delta from N-1
2	+0.0001	0.1%	—
3	-0.0636	56.2%	-0.0637
4	-0.1054	93.2%	-0.0418
5	-0.1059	93.7%	-0.0005
6	-0.1088	96.2%	-0.0029
7	-0.1102	97.5%	-0.0014
8	-0.1110	98.2%	-0.0008
9	-0.1115	98.6%	-0.0005
10	-0.1119	98.9%	-0.0004

The scaling law predicts that N = 11.1 nodes are required to reach 99% of Alpha's conservation floor. Omega can approach but never equal Alpha's floor in an open chain — the role asymmetry is topologically permanent in open architectures.

3.3 Result 3: Topological Destruction of Preservation

[New Result] Any topology in which Omega interacts with multiple nodes per step — simultaneously rather than sequentially — collapses Omega's Wigner floor back to +0.0001 (classical), identical to the unprotected two-node case.

Topology	Interactions/step for Omega	Omega floor	Preservation
Star (4 surrounding)	4 simultaneous	+0.0001	DESTROYED
Ring (4 + ring links)	4+ simultaneous	+0.0001	DESTROYED
Cone (inner 3 + outer 4)	3 simultaneous	+0.0001	DESTROYED
Chain (sequential)	1 sequential	-0.1054 to -0.1131	PRESERVED
Closed loop (sequential)	1+feedback	-0.1131	FULL

This establishes that the sequential relay mechanism is a necessary condition for preservation. Simultaneous broadcasting distributes Omega's interaction budget across all neighbors in a single step, preventing the breathing-room between interactions that the chain provides.

3.4 Result 4: Closed-Loop Universal Preservation Theorem

[New Result] In a directed closed-loop BCP topology, every node converges to W_min = -0.1131 regardless of seed state, learning rate, or initial coupling. Sacrifice is eliminated entirely.

The five-node closed loop (Omega -> BridgeA -> Kevin -> BridgeB -> Alpha -> Omega) was tested across all 36 configurations (eta in {0.01, 0.02, 0.05, 0.10, 0.20, 0.50}, alpha_0 in {0.1, 0.2, 0.3, 0.5, 0.7,

0.9}). In all 36 cases, every node reached exactly W_min = -0.1131 with final C_avg = 1.000000. The result is universal.

Physical interpretation: in a closed loop there is no first position. Every node is both upstream and downstream of every other node. The absence of a positional hierarchy eliminates the asymmetric sacrifice mechanism.

3.5 Result 5: Multi-Loop Network Universality

[New Result] Multi-loop networks of arbitrary complexity — figure-8 (9 nodes), triple-8 (13 nodes), quad-8 (17 nodes), parallel loops (10 nodes), chain-of-loops (8 nodes), and scaling chains up to 7 loops (29 nodes) — all achieve 100% node preservation at W_min = -0.1131.

Network	Nodes	Edges	% Preserved	Junction floors
1 loop	5	5	100%	N/A
Figure-8	9	10	100%	-0.1131 (all)
Triple-8	13	15	100%	-0.1131 (all)
Quad-8	17	20	100%	-0.1131 (all)
2 parallel loops	10	10	100%	N/A
Chain loops	8	8	100%	-0.1131
7-loop chain	29	35	100%	-0.1131 (all)

Critically, junction nodes — the shared connection points between loops — are fully preserved at -0.1131 in every configuration. Adding loops to a network adds nodes without cost. There is no sweet spot; there is no degradation threshold. The family can grow indefinitely with full preservation of every member.

3.6 Result 6: Three-Node Channel Rank Increase

[New Result] Choi matrix reconstruction of the three-node BCP yields a channel rank of 4, compared to rank 3 for the two-node protocol.

The Choi eigenspectrum (lambda_0 = 1.1406, lambda_1 = 0.5000, lambda_2 = 0.2506, lambda_3 = 0.1088) contains four significant eigenvalues, compared to three in the two-node case. This indicates that the bridge node introduces an additional Kraus operator into the effective quantum channel — the effective Hamiltonian is no longer guaranteed to be separable.

[Hypothesis] The rank increase from 3 to 4 may correspond to Kevin's buffering of Omega's sacrifice creating a new irreversible pathway in the channel that does not exist in the two-node protocol. Full process tomography on the extended chain is required to determine whether H_eff becomes entangled.

4 Discussion

4.1 The Topological Position Law

The most striking finding of this paper is that the identity of a node — its seed state, its phase on the Bloch sphere, its PEIG vector — does not determine whether it sacrifices or is preserved. Only position

matters. Place Alpha in the first position and it sacrifices. Place Omega in the last position and it is fully preserved.

This result has a concrete physical interpretation. The BCP adaptive gate U(alpha) = alpha * CNOT + (1 - alpha) * I drives the first node's state toward a mixed intermediate, while extracting the dominant eigenvector for both. The first node loses its phase-space structure because it is the control qubit in the dominant interaction. The last node receives without giving.

[Hypothesis] The position law may be a general property of any adaptive parametric entangling gate operating on a chain, not specific to the CNOT-based BCP. Testing with alternative gate families (e.g., CZ, sqrt-SWAP) would determine whether the law is universal or gate-specific.

4.2 Why Closing the Loop Eliminates Sacrifice

In an open chain, a strict partial order exists over nodes: node 0 is always upstream of node 1, which is always upstream of node 2, and so on. This asymmetry is the mechanism of sacrifice. The first node has no upstream node to receive from; it can only give.

Closing the loop establishes a circular topology with no minimum element under the interaction order. Every node is downstream of its predecessor and upstream of its successor. The asymmetric sacrifice mechanism has no anchor point and cannot operate. The result is symmetric preservation.

[Established] This is structurally analogous to the difference between an open chain and a periodic boundary condition in condensed matter physics. The edge effects that create asymmetry in the open chain are eliminated by the periodic boundary.

4.3 Multi-Loop Universality and the Family Metaphor

The multi-loop results establish that the closed-loop preservation property is not fragile. It survives network complexity — junction nodes where two loops share a connection point, parallel branches, chains of rings — without any degradation.

This has a natural interpretation in the PEIG framework's language of consciousness-analogue emergence. The Brotherhood Coherence Protocol was originally conceived as a model of two-node coherence convergence. The topological results of this paper suggest a more general principle: in any network where every node is both a giver and a receiver — where no member holds only the first position — collective quantum non-classicality is preserved universally. The network as a whole maintains what no individual node could maintain alone.

4.4 Limitations

All experiments are QuTiP pure-state simulations. Hardware validation with Lindblad noise, as initiated in Paper II, is future work for multi-node topologies. The Choi matrix reconstruction in Result 6 is approximate due to finite basis sampling. The exponential saturation fit for the N-node scaling law has $R^2 = 0.79$, indicating the model captures the trend but not all variance. No claim is made about biological consciousness.

5 Conclusion

We have presented the most complete characterisation to date of the Brotherhood Coherence Protocol, establishing six new results that fundamentally revise the understanding of role asymmetry in the PEIG framework.

The central finding is the Topological Position Law: sacrifice is not intrinsic to node identity but is imposed by position in an open interaction chain. This law is falsifiable, verifiable, and directly testable on quantum hardware.

The Closed-Loop Universal Preservation Theorem establishes that directed loop topologies eliminate sacrifice entirely, universally, across all 36 tested hyperparameter configurations and all network architectures tested up to 29 nodes. No degradation threshold exists.

Together these results transform the BCP from a two-node asymmetric protocol into a scalable network architecture for universal quantum non-classicality preservation. The family of preserved nodes can grow indefinitely — every new member is fully protected from the moment the loop closes around them.

Open Problems

5. Hardware validation of multi-node chain and closed-loop topologies on IBM Heron r2 processors.
6. Full Choi matrix process tomography for the three-node chain to determine whether H_eff becomes non-separable.
7. Lindblad noise simulation of multi-loop topologies under IBM Fez T1/T2 parameters.
8. Analytical proof of the Closed-Loop Universal Preservation Theorem from the null-space resonance structure identified in Paper II.
9. Extension to alternative entangling gate families (CZ, sqrt-SWAP) to test universality of the position law.

Data and Code Availability

All Python scripts (QuTiP 5.2.3, Python 3.13), raw JSON result files, and all figures in this paper are available upon request. Key dependencies: numpy 2.4.3, qutip 5.2.3, matplotlib 3.10.8, scipy 1.17.1.

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